

The Posterior Predictive Null

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Abstract. Bayesian model criticism is an important part of the practice of Bayesian statistics. Traditionally, model criticism methods have been based on the predictive check, an adaptation of goodness-of-fit testing to Bayesian modeling and an effective method to understand how well a model captures the distribution of the data. In modern practice, however, researchers iteratively build and develop many models, exploring a space of models to help solve the problem at hand. While classical predictive checks can help assess each one, they cannot help the researcher understand how the models relate to each other. This paper introduces the *posterior predictive null check* (PPN), a method for Bayesian model criticism that helps characterize the relationships between models. The idea behind the PPN is to check whether data from one model’s predictive distribution can pass a predictive check designed for another model. This form of criticism complements the classical predictive check by providing a comparative tool. A collection of PPNs, which we call a PPN study, can help us understand which models are equivalent and which models provide different perspectives on the data. With mixture models, we demonstrate how a PPN study, along with traditional predictive checks, can help select the number of components by the principle of parsimony. With probabilistic factor models, we demonstrate how a PPN study can help understand relationships between different classes of models, such as linear models and models based on neural networks. Finally, we analyze data from the literature on predictive checks to show how a PPN study can improve the practice of Bayesian model criticism. Code to replicate the results in this paper is available at <https://github.com/gemoran/ppn-code>.

Keywords: Predictive checks, Model criticism, Bayesian workflow.

1 Introduction

Bayesian model criticism is a crucial component of applied data analysis. While designing and studying Bayesian models, the goal of Bayesian model criticism is to understand in what ways the models fit the data well and in what ways they fall short.

One of the main tools for Bayesian model criticism is the *predictive check*, an adaptation of goodness-of-fit testing to Bayesian modeling (Box, 1980; Rubin, 1984; Meng, 1994; Gelman et al., 1996). Following the spirit of a goodness-of-fit test, a predictive check first sets a *reference* distribution, one that would have generated the data if the model was true. The check then asks whether the observed data—after applying a model-specific *diagnostic* function, such as a residual—could have plausibly arisen from

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that reference distribution. If the model passes this check, the observed data is said to be consistent with the model. If the check fails, it suggests that the model cannot adequately generate data similar to the observed data; as such, the model does not provide a useful representation of observed reality.

The most commonly used predictive check is the posterior predictive check (PPC) (Guttman, 1967; Rubin, 1984). A PPC sets its reference distribution to be the posterior predictive distribution of the data, and calculates the probability of the observed data (filtered through a diagnostic function) under this distribution. A PPC captures the idea that an adequate model is one whose posterior predictive provides a plausible distribution of the data.

Over the past decades, many researchers have innovated, refined, and expanded Bayesian predictive checks. Some work has explored the benefits of different reference distributions, such as the prior predictive (Box, 1980), the posterior predictive (Guttman, 1967; Rubin, 1984; Meng, 1994; Gelman et al., 1996), and combinations of the two (Evans and Moshonov, 2006). Other work considers different ways to assess the adequacy of the observed diagnostic, be it through a p -value, another measure of surprise (Bayarri and Berger, 1999), or a visual inspection (Gelman et al., 1996; Gelman, 2004). Still other work has addressed the problem of calibration, trying to ensure that the Bayesian model check enjoys sensible frequentist properties (Robins et al., 2000; Bayarri and Berger, 2000). Finally, there is a line of research that studies how to target the checks at specific components of the model (O’Hagan, 2003; Marshall and Spiegelhalter, 2007). This large body of work provides a rich toolbox for criticizing a Bayesian model.

However, there is an important side of model criticism that predictive checks do not address. In practice, rather than focus on a single model, most Bayesian researchers posit and criticize many models (Gelman et al., 2020). Given such a collection, predictive checks can assess each model individually, but they cannot compare the models to each other. Do some models capture different aspects of the data? Are some of them equivalent to each other?

To answer these questions, this paper proposes the *posterior predictive null check* (PPN). A PPN asks whether posterior predictive data from one model can pass the predictive check of another model. As an example, consider the simple data in Figure 1 (left): two-dimensional data from a mixture of *three* Gaussians. Given the data, Figure 1 (right) shows the posterior predictive distribution under four mixture models, with the number of mixture components $K \in \{1, 2, 3, 4\}$ (for details, see Appendix A of the Supplementary Material). As K increases, the posterior-predictive data looks more like the true distribution of the observed data; as expected, the posterior predictive for $K = 3$ is close to the truth. But notice the posterior predictive for $K = 4$ is equally good. While we might hope that a predictive check helps decide that $K = 1$ and $K = 2$ are inadequate, how can we detect that $K = 4$ offers no improvement over $K = 3$?

The PPN helps to solve this problem, asking *whether posterior data generated from the $K = 3$ posterior predictive passes the predictive check for $K = 4$* . As we will discuss, answering this question is equivalent to checking whether the predictive distribution for

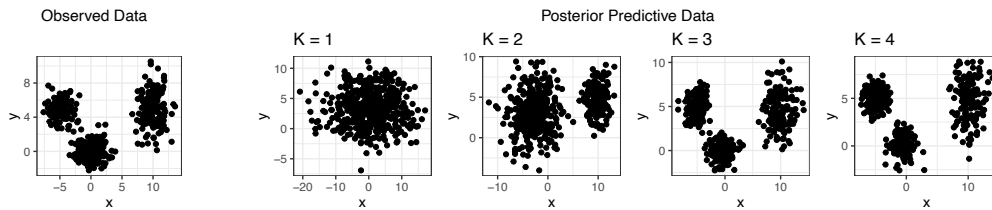


Figure 1: Posterior predictive draws from Gaussian mixture models of 2D data. On the far left is observed data $\mathbf{x}_{\text{obs}}$ from a Gaussian mixture with $K = 3$. Beside it are datasets drawn from the corresponding posterior predictive distributions of different mixtures $p_K(\mathbf{x}_{\text{rep}} | \mathbf{x}_{\text{obs}})$ with components $K \in \{1, 2, 3, 4\}$. Data drawn from $p_4(\mathbf{x}_{\text{rep}} | \mathbf{x}_{\text{obs}})$ (model-B posterior predictive) is indistinguishable from data drawn from $p_3(\mathbf{x}_{\text{rep}} | \mathbf{x}_{\text{obs}})$ (A-generated data). A PPN helps diagnose the fact $K = 4$ provides no improvement over $K = 3$. (This can help a researcher choose $K = 3$, for example, on the principle of parsimony.)

$K = 3$ is the same as the predictive distribution of $K = 4$. If model $K = 3$ passes this check, then $K = 3$ -generated data “fools” the posterior predictive check for $K = 4$; that is, the PPN suggests that $K = 4$ offers no additional modeling benefit, under the diagnostic used in the check.

We note that the name PPN comes from the idea that the “null distribution” of the PPC is the posterior predictive, and the principle of the PPC is that if the data came from the null then the model is consistent with the data. A PPN asks if an alternative model’s posterior predictive distribution might also produce the same null distribution.

For a set of models, a PPN study can help a researcher better understand the relationships between their models, both how they are redundant with each other and how they differ in their predictive distributions. As a demonstration, consider the matrix of plots in Figure 2. Each row indexes a model $K \in \{1, 2, 3, 4\}$. Along the diagonal are classical predictive checks—each panel illustrates the posterior predictive distribution of the model-specific diagnostic (here, a log likelihood) and the observed value. Notice here that all the models pass their predictive checks; each model can expand the variance of its components to capture the observed data. Consequently, these checks do not narrow down the set of models under consideration.

The PPNs in the off-diagonal panels of Figure 2 can help narrow down the set of models under consideration. Each PPN panel plots the distribution of the diagnostic under both the model under study (the row) and a simpler model (the column). When these distributions overlap then data from the column’s model can fool the check for the row’s model. We see that $K = 3$ cannot be fooled by $K = 2$ or $K = 1$; but $K = 4$, though consistent with the data, is fooled by $K = 3$. When working with an index of complexity—as we are for mixtures—the classical PPC helps indicate if a model’s complexity is sufficient to represent the observed data, while the PPN helps to determine whether that complexity is necessary to represent the data. (In Section 3 we will also study collections of models that are not indexed by complexity.)

We have demonstrated how a PPN study, by appealing to the concept of parsimony,